LOW RTO, NO PACKET LOSS, SIMPLE WINDOW ADVERTISEMENT

```
Select Flow Control Mode:
1. Window Advertising Mode (with Probing)
2. Explicit PAUSE/RESUME Mode
3. Packet Loss Mode (drop every 3rd packet)
Choice: 1
Receiver listening on port 8080...
[Handshake] Handshake Complete!

[Receiver] Received Packet SEQ: 0
[Receiver] Sent Cumulative ACK: 1 (Available Win:
9)
[Writer] Flushed SEQ 0 to disk.
[Receiver] Received Packet SEQ: 1
[Receiver] Sent Cumulative ACK: 2 (Available Win:
9)
[Receiver] Received Packet SEQ: 2
[Receiver] Sent Cumulative ACK: 3 (Available Win:
8)
[Writer] Flushed SEQ 1 to disk.
[Receiver] Received Packet SEQ: 3
[Receiver] Sent Cumulative ACK: 4 (Available Win:
8)
[Receiver] Received Packet SEQ: 4
[Receiver] Sent Cumulative ACK: 5 (Available Win:
7)
[Receiver] Received Packet SEQ: 5
[Receiver] Sent Cumulative ACK: 6 (Available Win:
6)
[Writer] Flushed SEQ 2 to disk.
[Receiver] Received Packet SEQ: 6
[Receiver] Sent Cumulative ACK: 7 (Available Win:
6)
[Writer] Flushed SEQ 3 to disk.
[Writer] Flushed SEQ 4 to disk.
```

```
Select Flow Control Mode:
1. Window Advertising Mode (with Probing)
2. Explicit PAUSE/RESUME Mode
3. Packet Loss Mode (receiver drops every 3rd pack
et)
Choice: 1
Enter RTO Value in ms (default 500): 300

--- Initiating Handshake ---
[Handshake] Connection Established!

[Chunker] Attached FLAG_EOF to packet sequence 6
[Chunker] Finished chunking file into 7 packets.
[Sender] Transmitted Packet SEQ: 0
[Sender] Transmitted Packet SEQ: 1
[Sender] Transmitted Packet SEQ: 2
[Sender] Transmitted Packet SEQ: 3
[Sender] Transmitted Packet SEQ: 4
[Sender] Transmitted Packet SEQ: 5
[Sender] Transmitted Packet SEQ: 6
[ACK Listener] Cumulative ACK: 1 (Advertised Win:
9)
[ACK Listener] Cumulative ACK: 2 (Advertised Win:
9)
[ACK Listener] Cumulative ACK: 3 (Advertised Win:
8)
[ACK Listener] Cumulative ACK: 4 (Advertised Win:
8)
[ACK Listener] Cumulative ACK: 5 (Advertised Win:
7)
[ACK Listener] Cumulative ACK: 6 (Advertised Win:
6)
[ACK Listener] Cumulative ACK: 7 (Advertised Win:
6)
```

```
[Receiver] Sent Cumulative ACK: 5 (Available Win:
7)
[Receiver] Received Packet SEQ: 5
[Receiver] Sent Cumulative ACK: 6 (Available Win:
6)
[Writer] Flushed SEQ 2 to disk.
[Receiver] Received Packet SEQ: 6
[Receiver] Sent Cumulative ACK: 7 (Available Win:
6)
[Writer] Flushed SEQ 3 to disk.
[Writer] Flushed SEQ 4 to disk.
[Receiver] Received FIN SEQ: 7 (still flushing buf
fer)
[Writer] Flushed SEQ 5 to disk.
[Writer] Flushed SEQ 6 to disk.
[Writer] Received EOF flag. Transfer complete.

--- Saved to received_payload.xlsx ---

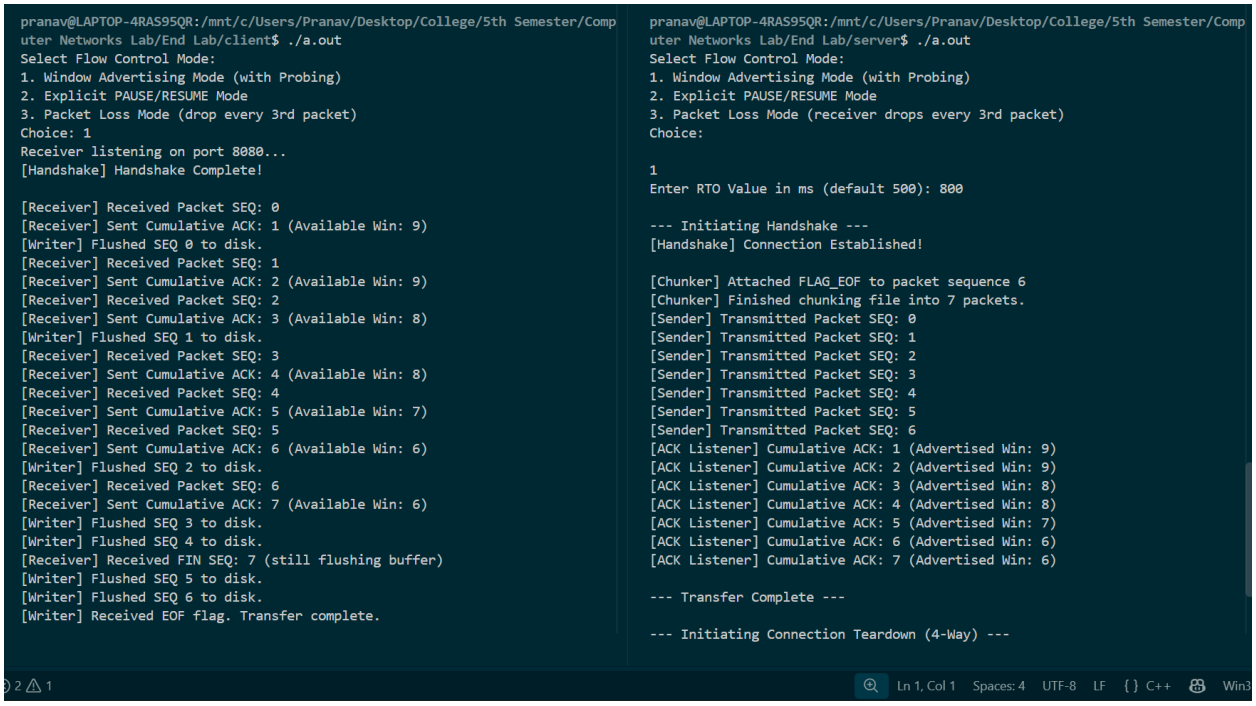
--- Connection Teardown (4-Way) ---
[Receiver] Sent ACK for FIN: 8
[Receiver State] ESTABLISHED -> CLOSE_WAIT
[Receiver] Application closing its side of the con
nection...
[Receiver] Sent FIN SEQ: 0
[Receiver State] CLOSE_WAIT -> LAST_ACK
[Receiver] Received ACK for FIN: 1
[Receiver State] LAST_ACK -> CLOSED
[Teardown] Connection Closed!
pranav@LAPTOP-4RAS95QR:/mnt/c/Users/Pranav/Desktop
/College/5th Semester/Computer Networks Lab/End La
b/client$
```

```
[Sender] Transmitted Packet SEQ: 5
[Sender] Transmitted Packet SEQ: 6
[ACK Listener] Cumulative ACK: 1 (Advertised Win:
9)
[ACK Listener] Cumulative ACK: 2 (Advertised Win:
9)
[ACK Listener] Cumulative ACK: 3 (Advertised Win:
8)
[ACK Listener] Cumulative ACK: 4 (Advertised Win:
8)
[ACK Listener] Cumulative ACK: 5 (Advertised Win:
7)
[ACK Listener] Cumulative ACK: 6 (Advertised Win:
6)
[ACK Listener] Cumulative ACK: 7 (Advertised Win:
6)

--- Transfer Complete ---

--- Initiating Connection Teardown (4-Way) ---
[Sender] Transmitted FIN SEQ: 7
[Sender State] ESTABLISHED -> FIN_WAIT_1
[ACK Listener] Received ACK for FIN: 8
[Sender State] FIN_WAIT_1 -> FIN_WAIT_2
[ACK Listener] Received FIN SEQ: 0 from receiver
[Sender] Transmitted ACK for FIN: 1
[Sender State] FIN_WAIT_2 -> TIME_WAIT
[Teardown] Waiting 2*MSL (2000 ms) in TIME_WAIT...
[Sender State] TIME_WAIT -> CLOSED
[Teardown] Connection Closed!
pranav@LAPTOP-4RAS95QR:/mnt/c/Users/Pranav/Desktop
/College/5th Semester/Computer Networks Lab/End La
b/server$
```

HIGH RTO, NO PACKET LOSS, SIMPLE WINDOW ADVERTISEMENT



```
pranav@LAPTOP-4RAS95QR:/mnt/c/Users/Pranav/Desktop/College/5th Semester/Comp
uter Networks Lab/End Lab/client$ ./a.out
Select Flow Control Mode:
1. Window Advertising Mode (with Probing)
2. Explicit PAUSE/RESUME Mode
3. Packet Loss Mode (drop every 3rd packet)
Choice: 1
Receiver listening on port 8080...
[Handshake] Handshake Complete!

[Receiver] Received Packet SEQ: 0
[Receiver] Sent Cumulative ACK: 1 (Available Win: 9)
[Writer] Flushed SEQ 0 to disk.
[Receiver] Received Packet SEQ: 1
[Receiver] Sent Cumulative ACK: 2 (Available Win: 9)
[Receiver] Received Packet SEQ: 2
[Receiver] Sent Cumulative ACK: 3 (Available Win: 8)
[Writer] Flushed SEQ 1 to disk.
[Receiver] Received Packet SEQ: 3
[Receiver] Sent Cumulative ACK: 4 (Available Win: 8)
[Receiver] Received Packet SEQ: 4
[Receiver] Sent Cumulative ACK: 5 (Available Win: 7)
[Receiver] Received Packet SEQ: 5
[Receiver] Sent Cumulative ACK: 6 (Available Win: 6)
[Writer] Flushed SEQ 2 to disk.
[Receiver] Received Packet SEQ: 6
[Receiver] Sent Cumulative ACK: 7 (Available Win: 6)
[Writer] Flushed SEQ 3 to disk.
[Writer] Flushed SEQ 4 to disk.
[Receiver] Received FIN SEQ: 7 (still flushing buffer)
[Writer] Flushed SEQ 5 to disk.
[Writer] Flushed SEQ 6 to disk.
[Writer] Received EOF flag. Transfer complete.
```

```
pranav@LAPTOP-4RAS95QR:/mnt/c/Users/Pranav/Desktop/College/5th Semester/Comp
uter Networks Lab/End Lab/server$ ./a.out
Select Flow Control Mode:
1. Window Advertising Mode (with Probing)
2. Explicit PAUSE/RESUME Mode
3. Packet Loss Mode (receiver drops every 3rd packet)
Choice:

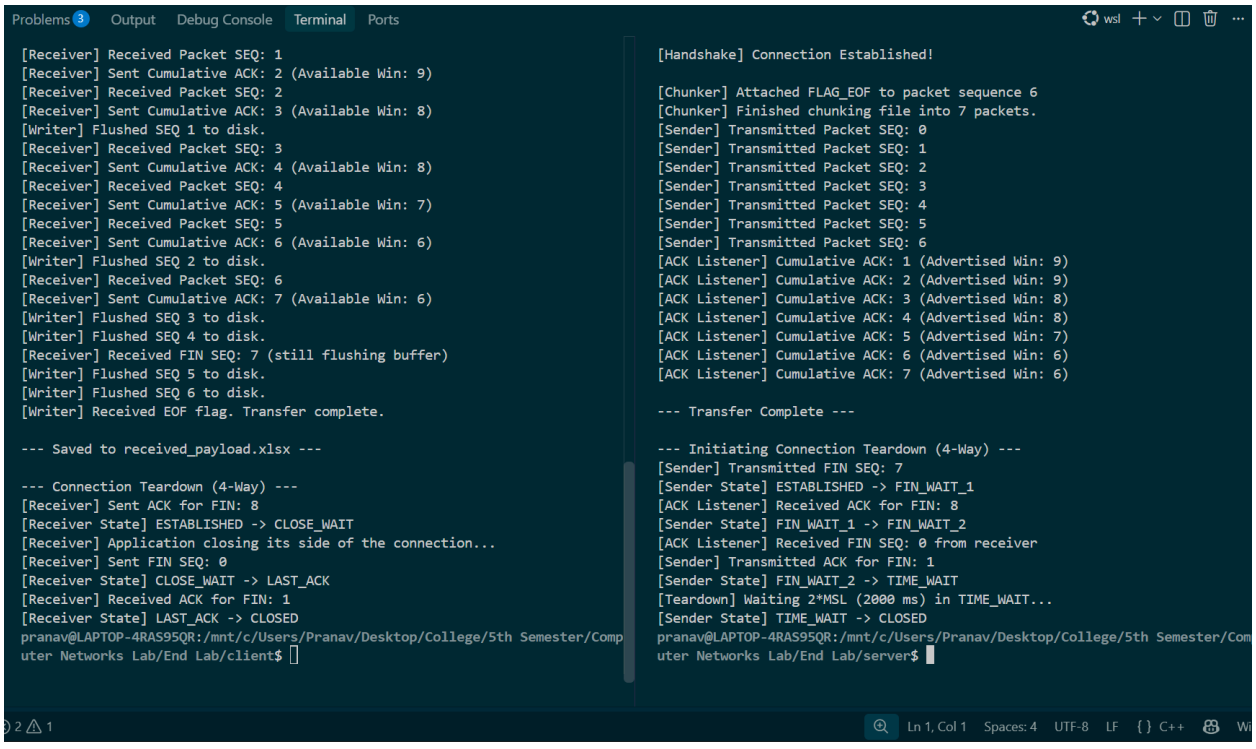
1
Enter RTO Value in ms (default 500): 800

--- Initiating Handshake ---
[Handshake] Connection Established!

[Chunker] Attached FLAG_EOF to packet sequence 6
[Chunker] Finished chunking file into 7 packets.
[Sender] Transmitted Packet SEQ: 0
[Sender] Transmitted Packet SEQ: 1
[Sender] Transmitted Packet SEQ: 2
[Sender] Transmitted Packet SEQ: 3
[Sender] Transmitted Packet SEQ: 4
[Sender] Transmitted Packet SEQ: 5
[Sender] Transmitted Packet SEQ: 6
[ACK Listener] Cumulative ACK: 1 (Advertised Win: 9)
[ACK Listener] Cumulative ACK: 2 (Advertised Win: 9)
[ACK Listener] Cumulative ACK: 3 (Advertised Win: 8)
[ACK Listener] Cumulative ACK: 4 (Advertised Win: 8)
[ACK Listener] Cumulative ACK: 5 (Advertised Win: 7)
[ACK Listener] Cumulative ACK: 6 (Advertised Win: 6)
[ACK Listener] Cumulative ACK: 7 (Advertised Win: 6)

--- Transfer Complete ---

--- Initiating Connection Teardown (4-Way) ---
```



```
Problems 3  Output  Debug Console  Terminal  Ports                    wsl

[Receiver] Received Packet SEQ: 1
[Receiver] Sent Cumulative ACK: 2 (Available Win: 9)
[Receiver] Received Packet SEQ: 2
[Receiver] Sent Cumulative ACK: 3 (Available Win: 8)
[Writer] Flushed SEQ 1 to disk.
[Receiver] Received Packet SEQ: 3
[Receiver] Sent Cumulative ACK: 4 (Available Win: 8)
[Receiver] Received Packet SEQ: 4
[Receiver] Sent Cumulative ACK: 5 (Available Win: 7)
[Receiver] Received Packet SEQ: 5
[Receiver] Sent Cumulative ACK: 6 (Available Win: 6)
[Writer] Flushed SEQ 2 to disk.
[Receiver] Received Packet SEQ: 6
[Receiver] Sent Cumulative ACK: 7 (Available Win: 6)
[Writer] Flushed SEQ 3 to disk.
[Writer] Flushed SEQ 4 to disk.
[Receiver] Received FIN SEQ: 7 (still flushing buffer)
[Writer] Flushed SEQ 5 to disk.
[Writer] Flushed SEQ 6 to disk.
[Writer] Received EOF flag. Transfer complete.

--- Saved to received_payload.xlsx ---

--- Connection Teardown (4-Way) ---
[Receiver] Sent ACK for FIN: 8
[Receiver State] ESTABLISHED -> CLOSE_WAIT
[Receiver] Application closing its side of the connection...
[Receiver] Sent FIN SEQ: 0
[Receiver State] CLOSE_WAIT -> LAST_ACK
[Receiver] Received ACK for FIN: 1
[Receiver State] LAST_ACK -> CLOSED
pranav@LAPTOP-4RAS95QR:/mnt/c/Users/Pranav/Desktop/College/5th Semester/Comp
uter Networks Lab/End Lab/client$
```

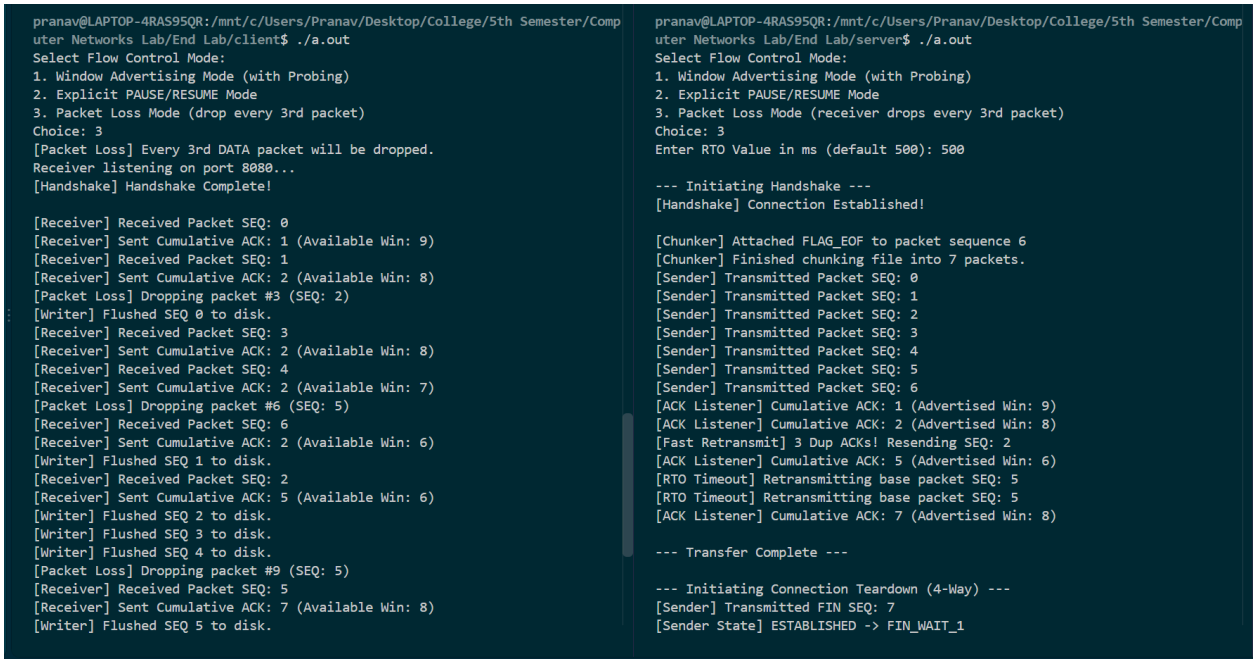
```
[Handshake] Connection Established!

[Chunker] Attached FLAG_EOF to packet sequence 6
[Chunker] Finished chunking file into 7 packets.
[Sender] Transmitted Packet SEQ: 0
[Sender] Transmitted Packet SEQ: 1
[Sender] Transmitted Packet SEQ: 2
[Sender] Transmitted Packet SEQ: 3
[Sender] Transmitted Packet SEQ: 4
[Sender] Transmitted Packet SEQ: 5
[Sender] Transmitted Packet SEQ: 6
[ACK Listener] Cumulative ACK: 1 (Advertised Win: 9)
[ACK Listener] Cumulative ACK: 2 (Advertised Win: 9)
[ACK Listener] Cumulative ACK: 3 (Advertised Win: 8)
[ACK Listener] Cumulative ACK: 4 (Advertised Win: 8)
[ACK Listener] Cumulative ACK: 5 (Advertised Win: 7)
[ACK Listener] Cumulative ACK: 6 (Advertised Win: 6)
[ACK Listener] Cumulative ACK: 7 (Advertised Win: 6)

--- Transfer Complete ---

--- Initiating Connection Teardown (4-Way) ---
[Sender] Transmitted FIN SEQ: 7
[Sender State] ESTABLISHED -> FIN_WAIT_1
[ACK Listener] Received ACK for FIN: 8
[Sender State] FIN_WAIT_1 -> FIN_WAIT_2
[ACK Listener] Received FIN SEQ: 0 from receiver
[Sender] Transmitted ACK for FIN: 1
[Sender State] FIN_WAIT_2 -> TIME_WAIT
[Teardown] Waiting 2*MSL (2000 ms) in TIME_WAIT...
[Sender State] TIME_WAIT -> CLOSED
pranav@LAPTOP-4RAS95QR:/mnt/c/Users/Pranav/Desktop/College/5th Semester/Com
uter Networks Lab/End Lab/server$
```

DEFAULT RTO, PACKET LOSS, SIMPLE WINDOW ADVERTISMENT



```
pranav@LAPTOP-4RAS95QR:/mnt/c/Users/Pranav/Desktop/College/5th Semester/Comp
uter Networks Lab/End Lab/client$ ./a.out
Select Flow Control Mode:
1. Window Advertising Mode (with Probing)
2. Explicit PAUSE/RESUME Mode
3. Packet Loss Mode (drop every 3rd packet)
Choice: 3
[Packet Loss] Every 3rd DATA packet will be dropped.
Receiver listening on port 8080...
[Handshake] Handshake Complete!

[Receiver] Received Packet SEQ: 0
[Receiver] Sent Cumulative ACK: 1 (Available Win: 9)
[Receiver] Received Packet SEQ: 1
[Receiver] Sent Cumulative ACK: 2 (Available Win: 8)
[Packet Loss] Dropping packet #3 (SEQ: 2)
[Writer] Flushed SEQ 0 to disk.
[Receiver] Received Packet SEQ: 3
[Receiver] Sent Cumulative ACK: 2 (Available Win: 8)
[Receiver] Received Packet SEQ: 4
[Receiver] Sent Cumulative ACK: 2 (Available Win: 7)
[Packet Loss] Dropping packet #6 (SEQ: 5)
[Receiver] Received Packet SEQ: 6
[Receiver] Sent Cumulative ACK: 2 (Available Win: 6)
[Writer] Flushed SEQ 1 to disk.
[Receiver] Received Packet SEQ: 2
[Receiver] Sent Cumulative ACK: 5 (Available Win: 6)
[Writer] Flushed SEQ 2 to disk.
[Writer] Flushed SEQ 3 to disk.
[Writer] Flushed SEQ 4 to disk.
[Packet Loss] Dropping packet #9 (SEQ: 5)
[Receiver] Received Packet SEQ: 5
[Receiver] Sent Cumulative ACK: 7 (Available Win: 8)
[Writer] Flushed SEQ 5 to disk.
```

```
pranav@LAPTOP-4RAS95QR:/mnt/c/Users/Pranav/Desktop/College/5th Semester/Comp
uter Networks Lab/End Lab/server$ ./a.out
Select Flow Control Mode:
1. Window Advertising Mode (with Probing)
2. Explicit PAUSE/RESUME Mode
3. Packet Loss Mode (receiver drops every 3rd packet)
Choice: 3
Enter RTO Value in ms (default 500): 500

--- Initiating Handshake ---
[Handshake] Connection Established!

[Chunker] Attached FLAG_EOF to packet sequence 6
[Chunker] Finished chunking file into 7 packets.
[Sender] Transmitted Packet SEQ: 0
[Sender] Transmitted Packet SEQ: 1
[Sender] Transmitted Packet SEQ: 2
[Sender] Transmitted Packet SEQ: 3
[Sender] Transmitted Packet SEQ: 4
[Sender] Transmitted Packet SEQ: 5
[Sender] Transmitted Packet SEQ: 6
[ACK Listener] Cumulative ACK: 1 (Advertised Win: 9)
[ACK Listener] Cumulative ACK: 2 (Advertised Win: 8)
[Fast Retransmit] 3 Dup ACKs! Resending SEQ: 2
[ACK Listener] Cumulative ACK: 5 (Advertised Win: 6)
[RTO Timeout] Retransmitting base packet SEQ: 5
[RTO Timeout] Retransmitting base packet SEQ: 5
[ACK Listener] Cumulative ACK: 7 (Advertised Win: 8)

--- Transfer Complete ---

--- Initiating Connection Teardown (4-Way) ---
[Sender] Transmitted FIN SEQ: 7
[Sender State] ESTABLISHED -> FIN_WAIT_1
```

```
[Receiver] Sent Cumulative ACK: 2 (Available Win: 8)
[Receiver] Received Packet SEQ: 4
[Receiver] Sent Cumulative ACK: 2 (Available Win: 7)
[Packet Loss] Dropping packet #6 (SEQ: 5)
[Receiver] Received Packet SEQ: 6
[Receiver] Sent Cumulative ACK: 2 (Available Win: 6)
[Writer] Flushed SEQ 1 to disk.
[Receiver] Received Packet SEQ: 2
[Receiver] Sent Cumulative ACK: 5 (Available Win: 6)
[Writer] Flushed SEQ 2 to disk.
[Writer] Flushed SEQ 3 to disk.
[Writer] Flushed SEQ 4 to disk.
[Packet Loss] Dropping packet #9 (SEQ: 5)
[Receiver] Received Packet SEQ: 5
[Receiver] Sent Cumulative ACK: 7 (Available Win: 8)
[Writer] Flushed SEQ 5 to disk.
[Writer] Flushed SEQ 6 to disk.
[Writer] Received EOF flag. Transfer complete.
[Receiver] Received FIN SEQ: 7 (still flushing buffer)

--- Saved to received_payload.xlsx ---
[Packet Loss] Dropped 3 of 10 DATA packets received.

--- Connection Teardown (4-Way) ---
[Receiver] Sent ACK for FIN: 8
[Receiver State] ESTABLISHED -> CLOSE_WAIT
[Receiver] Application closing its side of the connection...
[Receiver] Sent FIN SEQ: 0
[Receiver State] CLOSE_WAIT -> LAST_ACK
[Receiver] Received ACK for FIN: 1
[Receiver State] LAST_ACK -> CLOSED
[Teardown] Connection Closed!
pranav@LAPTOP-4RAS95QR:/mnt/c/Users/Pranav/Desktop/College/5th Semester/Comp
uter Networks Lab/End Lab/client$
```

```
[Handshake] Connection Established!

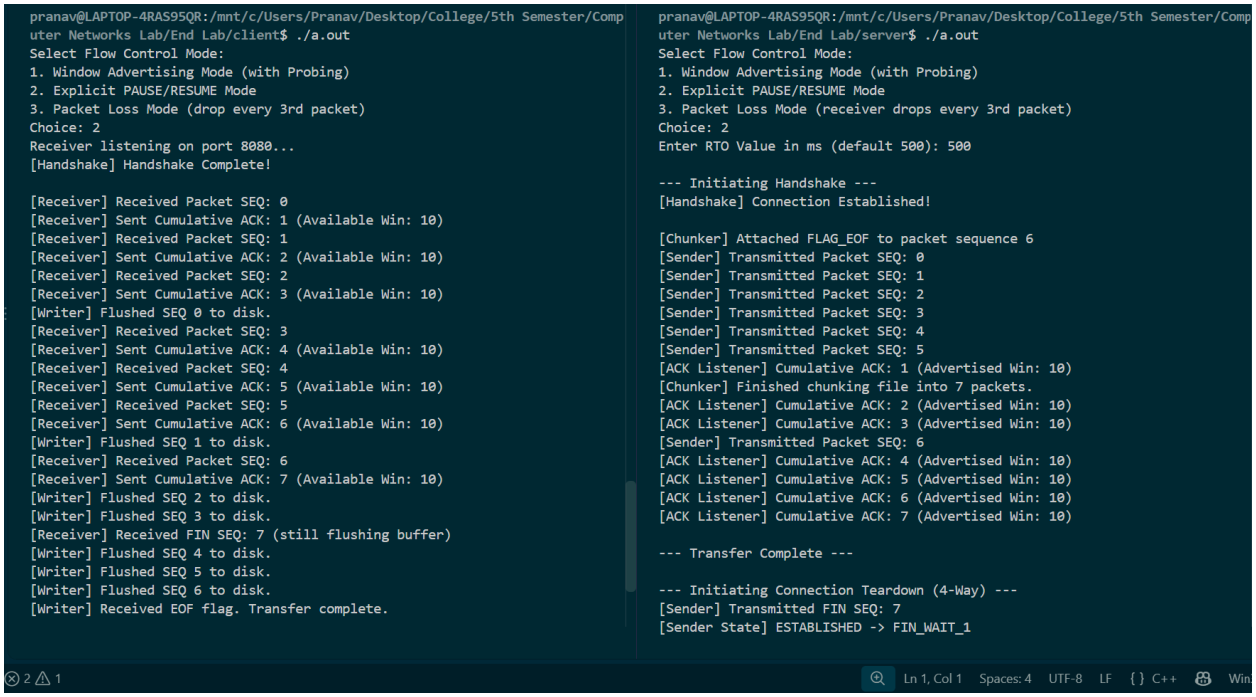
[Chunker] Attached FLAG_EOF to packet sequence 6
[Chunker] Finished chunking file into 7 packets.
[Sender] Transmitted Packet SEQ: 0
[Sender] Transmitted Packet SEQ: 1
[Sender] Transmitted Packet SEQ: 2
[Sender] Transmitted Packet SEQ: 3
[Sender] Transmitted Packet SEQ: 4
[Sender] Transmitted Packet SEQ: 5
[Sender] Transmitted Packet SEQ: 6
[ACK Listener] Cumulative ACK: 1 (Advertised Win: 9)
[ACK Listener] Cumulative ACK: 2 (Advertised Win: 8)
[Fast Retransmit] 3 Dup ACKs! Resending SEQ: 2
[ACK Listener] Cumulative ACK: 5 (Advertised Win: 6)
[RTO Timeout] Retransmitting base packet SEQ: 5
[RTO Timeout] Retransmitting base packet SEQ: 5
[ACK Listener] Cumulative ACK: 7 (Advertised Win: 8)

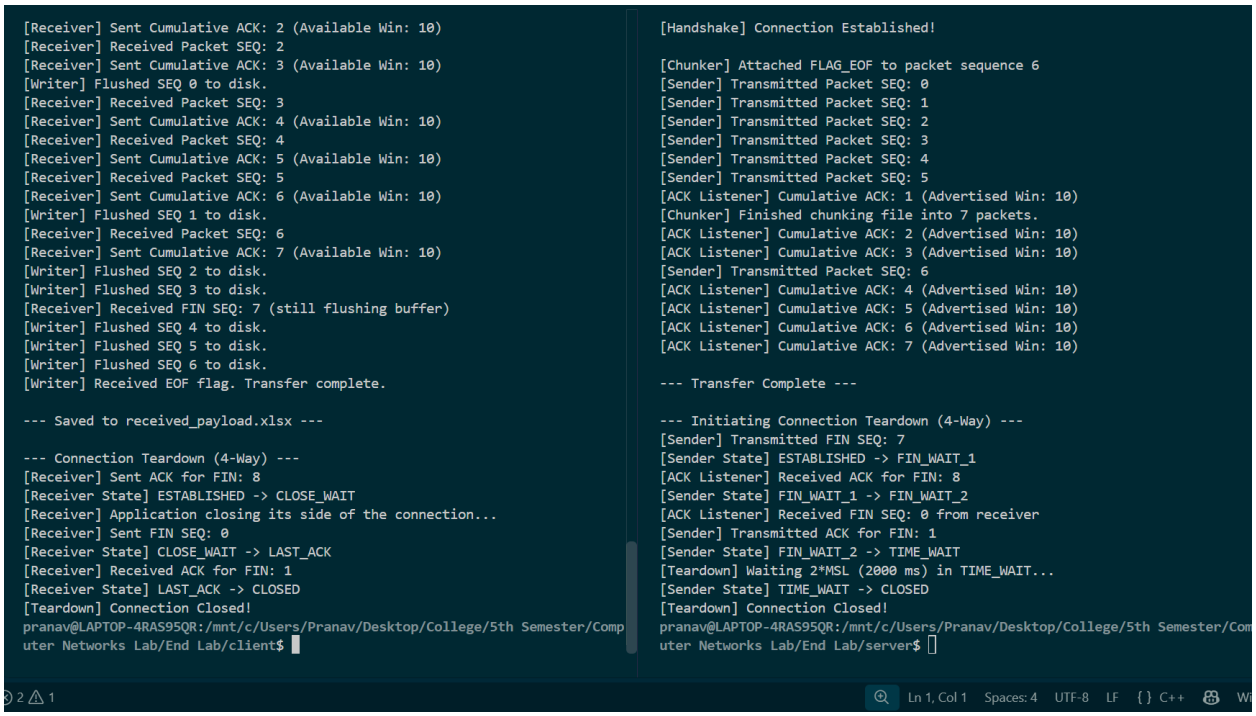
--- Transfer Complete ---

--- Initiating Connection Teardown (4-Way) ---
[Sender] Transmitted FIN SEQ: 7
[Sender State] ESTABLISHED -> FIN_WAIT_1
[ACK Listener] Received ACK for FIN: 8
[Sender State] FIN_WAIT_1 -> FIN_WAIT_2
[ACK Listener] Received FIN SEQ: 0 from receiver
[Sender] Transmitted ACK for FIN: 1
[Sender State] FIN_WAIT_2 -> TIME_WAIT
[Teardown] Waiting 2*MSL (2000 ms) in TIME_WAIT...
[Sender State] TIME_WAIT -> CLOSED
[Teardown] Connection Closed!
pranav@LAPTOP-4RAS95QR:/mnt/c/Users/Pranav/Desktop/College/5th Semester/Comp
uter Networks Lab/End Lab/server$
```

⊗ 2 ⚠ 1 Ln 1, Col 1 Spaces: 4 UTF-8 LF { } C++ Win32

DEFAULT RTO, NO PACKET LOSS, PAUSE/RESUME





Implemented pause/resume for congestion control, but file is very small